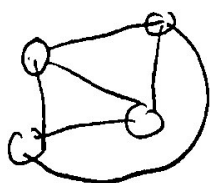


Theorem: An undirected graph has even number of vertices of odd degree.

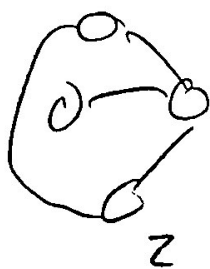
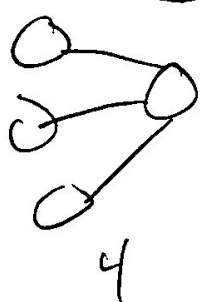
Proof: $2m = \sum_{v \in V} \deg(v)$ by Handshaking Lem.

assume only odd degree vertices exist



If $\exists$ a vertex of odd degree

$$\text{Total edges} = \sum_{v \in V} \deg(v)$$



$$e = \frac{\sum_{v \in V} \deg(v)}{2}$$

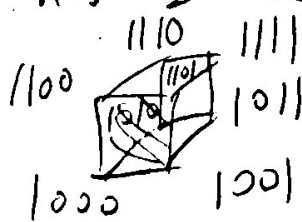
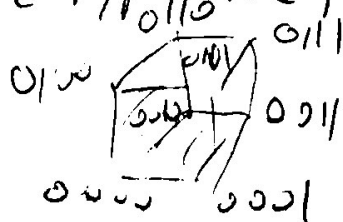
$$e \pmod{2} = 0$$

$$\therefore e = 0 \pmod{2}$$

$\therefore$ If $\exists$ an odd $\deg(v_i)$, there must be another different vertex of odd degree. The pairs are independently grouped, or required an even number of odd degreed vertices, i.e. $\sum_{v_j \in V} \deg(v_j)$, v_j are odd degreed

$$\sum_{v_j \in V} \deg(v_j) = 0 \pmod{2} \Rightarrow \deg(v_j) = 0 \pmod{2}$$

4-d hypercube, Q_4 has 2^4 vertices



respective

& shaded faces

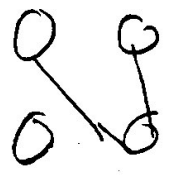
can join those enclosure,

Theorem: A simple graph is bipartite iff possible to assign one of two diff colours to each vertex s.t. no 2 adjacent vertices have same colour

Proof: A bipartite graph has s.t. $N(v_1)$ neighbours are disjoint coloured. Hence, 2 colours for each disjoint partitioned subset V_1, V_2 .

Theorem: Hall's Marriage Theorem complete matching iff
 $|N(A)| \geq |A| \quad \forall A \subseteq V_1$

Proof: $G = (V, E)$, bipartition (V_1, V_2)

V_1  $|N(A)| = 1, |A| = 2 \quad 1 \not\geq 2$

let $|N(A)| = 0 \Rightarrow |N(A)| \not\geq |A| \quad \forall A \subseteq V_1$.

$\therefore |N(A)| \geq 1$

let $|N(A)| = 1 \Rightarrow |A| = 1$

let $|N(A)| = 2 \Rightarrow |A| \leq 2$

let $|N(A)| = n \Rightarrow |A| \leq n$

$\Rightarrow |N(A)| \geq |A|$

~~let~~ $|N(A)| \neq 0$

$|N(A)| = 1 \Rightarrow |A| = 1$

and complete matching of subset A

- base case.

Theorem: There is a simple path between every pair of distinct vertices of a connected undirected graph.

proof: let v_1, v_2 denote the two distinct vertices,

By connectedness, $\exists$ a path between (from) v_1 to v_2 .

Prove simplicity, no repeating edges.

By connectedness, this implies an existence of a simple path between 2 vertices,

Assume path is not simple, the ~~repeat~~ repeating edge circuit can be removed!

$$K(G) \leq \lambda(G) \leq \min_{v \in V} \deg(v)$$

proof:

(Ex 17, 65)

final

~~$K(G) \leq \lambda(G)$: an isolation can only occur, if~~

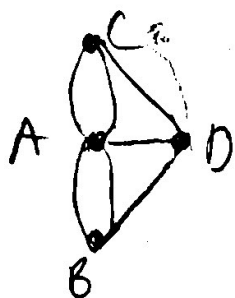
~~$K(G) \leq \max_{v \in V} \deg(v)$: removing $\max_{v \in V} \deg(v)$ results in a isolated node (vertex).~~

~~$\therefore K(G) \leq \min_{v \in V} \deg(v)$: $K(G)$ is defined to be the minimum vertex connectivity.~~

$$\Rightarrow \lambda(G) \leq \min_{v \in V} \deg(v).$$

~~$K(G) \leq \lambda(G)$ ~~iff~~ is clear if $\min_{v \in V}$ is a bridging vertex; which it is, since $K(G)$ and $\lambda(G)$ exist on $\min_{v \in V} \deg(v)$.~~

Seven bridges of Königsberg!



Euler circuit? There

No, due to having to start at $N(v) = 0 \pmod{2} \Rightarrow A$, and symmetry, we see that odd connection (1)

to D proves it not an Euler circuit.

Theorem:

$\Rightarrow$ even number of edges for all vertices, starting/ending, odd + 1 = even, odd + 1 = even

going to it.

odd + 1 = even, odd + 1 = even

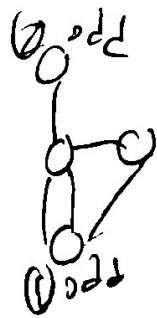
Euler path

① end point

② start point

Traveling salesman problem

Chinese Postman problem, ...



① odd: must return back to a nonstarting location, hence 1 odd,

② odd: must strictly leave

starting point, shortest circuit Hamiltonian vertex-based

shortest circuit Eulerian edge-based

vertices on dodecahedron = $5 \cdot 12 = 60$

$= 60 = 2 \cdot 12$

$24 - 2 \cdot 6$ #edges = #vertices

$\Rightarrow 5 + 5 \cdot 4 + 5 \cdot 3 + 6 =$

$\Rightarrow 5 + 4 + 4 + 3 + 3 + 2 + 10 =$

$\Rightarrow 5 + 4 + 3 + 3 + 2 + 3 + 3 + 2 + 1 = 32$

simple no. recs. & suff. = $5 + 6 + 9 + 4 + 6$

conditions, many suff. = 30

Hamilton Circuit: inspiration from dodecahedron:

- every vertices' degree should be odd, since ~~edge~~ an odd amount of edges won't be "used", just vertices?



it seems a Hamilton circuit exists for $\deg(v), v \in V \geq 1$ and

$N(v), v \in V \geq 1$, number of edges ~~disproves~~



- A Hamilton circuit can not contain a smaller circuit within it.

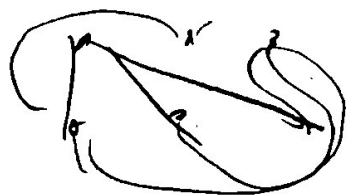
A Hamilton circuit doesn't have any bridge vertices that if removed form 2 connected?

Theorem: Dijkstra's algorithm is $O(n^2)$, $n = \# \text{ vertices}$

proof:

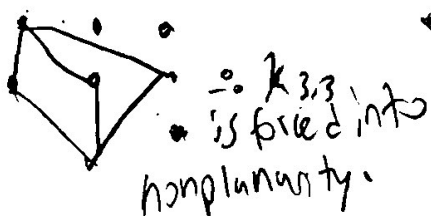
for a ^{simple} planar graph $G=(V,E)$

$K_{3,3}$ is non-planar Euler's Formula: e edges, v vertices
 $\Rightarrow r_{\text{regions}} = e - v + 2$



an edge adds 1 region

an vertex and 2 edges adds 1 region.



$$e - (v - 2)$$

$$= e - v + 2$$

Proof of Euler's Formula $r = e - v + 2$ ^{-adding}
 induction:

base case:

$$v=2, e=1 \Rightarrow r = e - v + 2 = 3 - 2 = 1$$

inductive step:

every increase in edge increases r by 1

$$r_{\text{+edge}} = (e+1) - v + 2$$

$$\Rightarrow r_{\text{+edge}} + 1 = (e+1) - v + 2 \quad \text{regions}$$

every increase in vertices and 2 edges increase by 1

$$r_{\text{+iv}} = (e+2) - (v+1) + 2$$

$$r_{\text{+iv}} = (e+2) - (v+1) + 2$$

$$r_{\text{+iv}} = e - v + 2 + 1$$

proof:

$$r_{\text{bc}} = e_1 + v_2 + 2 = 1$$

inductive step:

$$r_{\text{e+1}} = e_{i+1} - v_i + 2 = r_{e_i} + 1$$

$$r_{\text{v+1}} = e_{i+2} - v_{i+1} + 2 = r_{v_i} + 1$$

by liberality $\therefore r_{e_{i+1}, v_{i+1}} = e_{i+3} - v_{i+1} + 2 = r_{e_i, v_i} + 2$

Corollary 1: If G is a connected planar simple graph w/ e edges, v vertices,
 $v \geq 3$ then $e \leq 3v - 6$

proof: If G is complete $\Rightarrow \deg(v_i)_{i \in V} = \frac{n(n-1)}{2} = \frac{(v-1)}{2} \# \text{ edges}$
 - by handshake lemma.

$$\therefore e \leq v \frac{(v-1)}{2} \leq 3v - 6$$

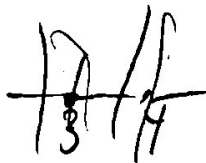
$$v^2 - v \leq 6v - 12$$

$$\Rightarrow v^2 - 7v + 12 \leq 0$$

$$\Rightarrow (v-3)(v-4) \leq 0$$

$$\Rightarrow 3 \leq v \leq 4 \text{ valid } \because K_5 \text{ is not planar, } i \geq 5$$

• So Euler's Formula proves K_5 is not planar.



$$\therefore e \leq 3v - 6$$

- prove that $K_i, i \geq 5$ is non planar

proof:



If a vertex is not adjacent to R_i ,
 the unbounded region, then G is non planar

- use regions to prove nonplanarity,

• A planar graph has forming regions @ vertices no intersections

- can prove by inspection that K_5 is non-planar $K_i, i \geq 5$ is non planar.

Textbook: $2e = \left[\sum_{r_i} \deg(r_i) \right] \geq 3r$ for connected, planar, simple graphs.
 $\Rightarrow \frac{2}{3}e \geq r$ each region has at least 3 edges.

$$\Rightarrow \frac{2}{3}e \geq e - v + 2$$

$$\Rightarrow e \leq 3v - 6. \text{ } \therefore K_5 \text{ is non-planar. } e = 10 \neq 3v - 6 = 9$$

$K_{3,3}$ satisfies

degree at region must be at least 4,

* proof. This if the

- If a connected, planar, simple graph has e edges & v vertices $v \geq 3$ and no circuit of length 3, $e \leq 2v - 4$, provable since $2e = \sum_{r_i} \deg(r_i) \geq 4r$

ex) $\chi(K_n) = n-1$

ex) $\chi(K_{m,n}) = 2$

ex) $\chi(C_n), n \geq 3 = 2$ if n even 3 if n odd

7) $K_{n_1, n_2, \dots, n_m}$ edges = ?

vertices is $n_1 + n_2 + \dots + n_m$

edges $n_1 \cdot n_2 + \dots + n_m$

$= n_1(n_2 + n_3 + \dots + n_m)$

$+ n_2(n_3 + n_4 + \dots + n_m) = n_i \cdot (N - n_i)$

$+ \dots$

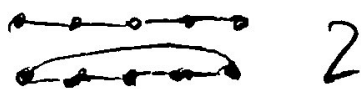
$+ n_{m-1} \cdot n_m$



$n_m \cdot (n_1 + n_2 + \dots + n_{m-1})$
 $+ n_{m-1} \cdot (n_1 + n_2 + \dots + n_{m-2})$
 $+ n_{m-2} \cdot (n_1 + \dots + n_{m-3})$

#V = $\sum_{i=1}^m \left[n_i \cdot \sum_{j=1}^{i-1} n_j \right]$

32 a) 5 vertices, non isomorphic simple connected graphs,



2

b) $\chi(G) = 4$, non isomorphic simple conn graph w/ 5 vertices

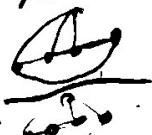
all graphs w/ 5 vertices s.t. $\deg(v) \leq 3 \forall v$.

$\deg \leq 2$ has 2, above, $\deg \leq 3$ has

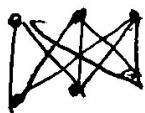
c) non planar

$1 + 2C_4 + 3C_5 + 2C_1 + 2C_6$

$+ 3C_1 + 2C_1 + 2C_6 - 2C_6$

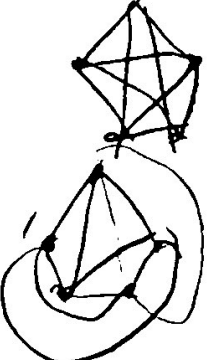
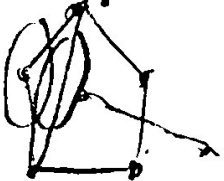


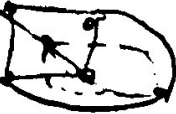
Kuratowski's Theorem: A graph is nonplanar iff it has a subgraph homeomorphic to $K_{3,3}$ or K_5 .

Proof: $K_{3,3}$  $\dots$

 $K_{3,3}$ represents the smallest unit with 2 internal regions
 $\{v_1, v_4\}$ or $\{v_2, v_3\}$ connected to v_5 ,
 v_5 connected to opposite pair, and a nonplanarity edge

- smallest unit ~~not~~ has nodes: $K_5, K_{3,3}$

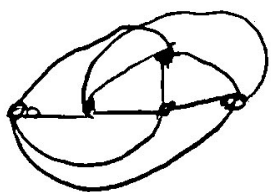
K_5  K_4  homeomorphic  

It is intuitive that  from K_5 and  from $K_{3,3}$ represent smallest ^{nonplanar} homeomorphic subgraphs since K_5 represents an enclosed region being crossed and $K_{3,3}$ represents an enclosed region having to be reached from the outside due to 2 regions on the inside. K_5 is simpler, $K_{3,3}$ can be extrapolated for n regions where an access to a middle point is restricted to an outside-region vertex, hence a crossover, proof by induction ??? for $K_{3,3}$ (and K_5 simpler), will continue the induction proof.



$$4[3^c_1 + 3^c_2] = 4[3 + 3] = 24$$

$$3, 1, 1, 2, 4+3+2+3+2+1 = 12+4 = 16$$



$$\# \text{ edges} = \frac{5 \cdot 4}{2} = 10 = 18$$

c) non planarity $\Rightarrow K_{3,3}, K_5$ embedded

$K_5 \Rightarrow 1$

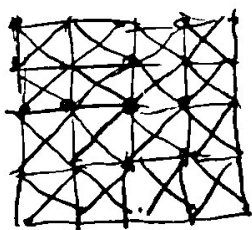
$K_{3,3}$



has 6 nodes - " 1

27) min. # queens controlling $n \times n$ chessboard.

$n=3$



1 Q



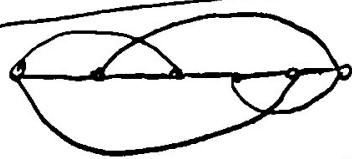
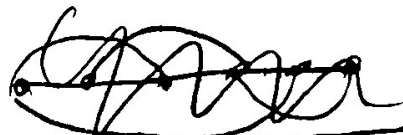
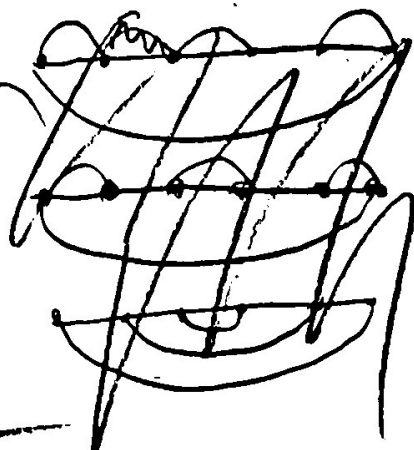
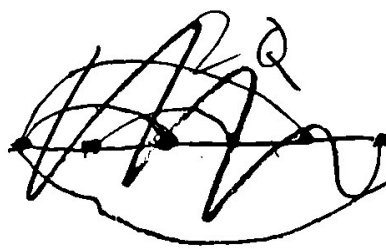
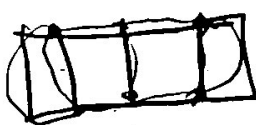
$n=4$

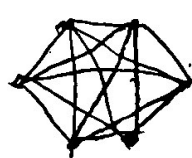
≤ 4 Q

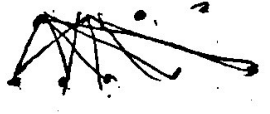
2 Q

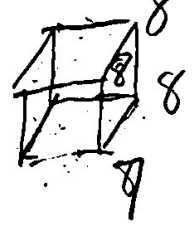
$n=5$

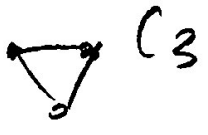
47)



52) a. K_6 $r_{K_6} = 6 = n$ K_6 
 $d_{K_6} = 6 = n$

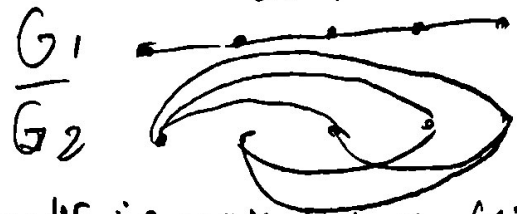
b. $K_{4,5}$ $r = 8 = n-1$ 
 $d = 8 = n-1$ 

c. Q_3 hypercube 2^3 
~~max edges~~ Path: $r = 7$
 $d = 8$

d. C_6 $r = 5 = 6-1$ 
 $d = 5 = 6-1$

53a) if $d \geq 4$ of simple graph G , $\bar{G}$ da $d \leq 2$

$d \geq 4 \Rightarrow \geq 4$ edges $\Rightarrow \geq 5$ vertices
 $d = 4$ if $d \geq 4 \Rightarrow \chi(G)$ decreases monotonically.



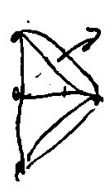
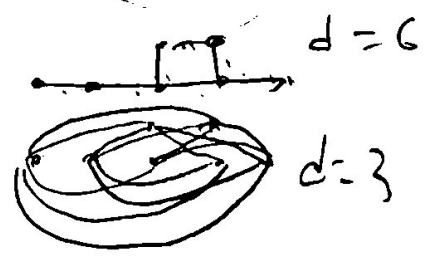
Diameter $G_2 = 2 \leq 2$

- diameter is max distance between 2 vertices.



53b)

54) multigraph w/ $2m$ vertices, ^{even} odd degree.
 Show every edge of a circuit ^{circuit w/ all edges} contains at least m edges more than once.



$m=2$ for ^{on vertex} every edge, there will be an extra input since odd degree. $\Rightarrow$ +1 repeated edge to leave.
 assume next edge traversal is not repeated $\therefore$
 $\frac{1}{2}$ repeated edges on minimum $\frac{1}{2} = \frac{2m}{2} = m$
 $\therefore$ repeated edges $\geq m$